

Crypto Comp — Launch & Growth Plan

Context

Crypto Comp (`com.simplesignllc.cryptocomp` , v1.3.4) is a polished, production-ready free-to-play **crypto trading-competition** app. The product and engagement layers are strong; the **user-acquisition layer is nearly empty**. This is a go-to-market strategy that (a) closes the measurement/virality gaps that would otherwise make any ad spend wasteful, then (b) drives retained users US-first on a lean budget.

Locked inputs: budget < \$500/mo, **XP-only** prizes at launch (keep `PAYOUTS_ENABLED` off), north-star metric **retained active users (D7/D30 / weekly active competitors)**, market **US-first**.

A companion build backlog with a ready-to-run Claude prompt lives in `../future_todo.md`.

What exists today (grounding facts)

- **Product:** \$100k paper bankroll; Daily/Featured/1v1/Replay contests; 6-hour Sprint (20 players) and Weekly (1000 players) auto-created by Lambda; leagues Bronze→Platinum, XP, quests, streaks, copy-trading, price-prediction game. Files: `src/screens/CompeteScreen.tsx` , `src/services/gamification.ts` , `src/constants/featureFlags.ts` .
 - **Monetization:** AdMob (banner/interstitial/rewarded/native, capped in `src/lib/adManager.ts`); RevenueCat IAP — No-Ads \$2.99/mo, Premium \$3.99/mo, \$5M balance \$2.99 (`src/components/PurchaseModal.tsx`); Stripe Connect cash payouts built but flag-gated off; default prize **5000 XP**.
 - **Engagement infra:** push notifications with audience segmentation (`src/services/pushDeviceService.ts` , notification-campaign Lambda); native Share for profile/duel/trade.
 - **Acquisition gaps (the problem):**
 - **No referral system** — only untracked 1v1 duel codes in `CompeteScreen.tsx` .
 - **No deep-link attribution** — shares are plain text; no Branch/universal links/URL scheme.
 - **No product analytics** — only Sentry error tracking; no funnel/retention/cohort events.
 - **Auth friction** — email/password only; no Apple Sign-In or guest mode (`src/screens/AuthScreen.tsx`).
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Phase 0 — Readiness (do BEFORE spending a dollar; ~1–2 weeks)

Spending on ads with zero attribution/analytics is the single biggest avoidable mistake here.

1. **Product analytics (free tier):** PostHog or Amplitude. Instrument the core funnel — install → signup → onboarding complete → first trade → first contest join → D1/D7 return. Hook into existing choke points: `OnboardingWalkthrough.tsx` , `TradeScreen.tsx` , `CompeteScreen.tsx` . This is what makes "retained active users" measurable.
2. **Attribution + deep links:** Branch (free <10k MAU) or Apple Search Ads + Apple's own attribution. Enables referral tracking AND paid-channel CAC. Add a URL scheme/universal link in `app.json` (none today).
3. **Cut signup friction:** add **Sign in with Apple** + optional **guest mode** so users can reach the first contest before creating an account (`AuthScreen.tsx` , Cognito). Biggest retention lever for a free app.
4. **ASO pass (free, highest ROI channel):** title/subtitle/keywords around "crypto trading game / simulator / contest / paper trading / fantasy crypto"; screenshots that lead with the **competition + leaderboard** hook, not a generic portfolio. Localize for US English.

Positioning & message

"**Fantasy sports for crypto.**" Free, everyone starts with the same \$100k, best P&L wins — no real money at risk. Lead every asset with **competition + leaderboard + bragging rights**, not "practice trading" (crowded, low-emotion). Hooks: 6-hour sprints (fast dopamine), 1v1 duels (built-in 1:1 virality), seasonal leagues (retention).

Plan A — Where to advertise (US-first, lean < \$500/mo)

Lean budget → **one paid channel done well + heavy organic**. Recommended monthly split:

Channel	Spend	Why
Apple Search Ads (exact-match: brand + high-intent e.g. "crypto trading game", "paper trading", "crypto simulator")	\$250–300	Highest-intent, cheapest measurable installs for iOS; tight US targeting; works at small budgets. Primary paid channel.
Creative testing — TikTok Spark Ads <i>or</i> Reddit Ads (r/CryptoCurrency-adjacent), boosting your best organic video	\$100–150	Find one winning creative before ever scaling. Treat as research, not growth.
Micro-influencer / community seeding (gift Premium codes, sponsor a community contest)	\$0–100	Often free via affiliate/XP; see referral + free playbook.

Rule: do not increase spend until Apple Search Ads CAC is known and D7 retention $\geq$ target. Gate scaling on measured CAC < projected LTV (ad ARPDAU + IAP), not on install count.

Free / organic user acquisition (the main engine on this budget)

1. **Build-in-public on X/Twitter:** daily/weekly leaderboard screenshots, biggest-winner P&L, "this week's champion." Crypto X loves competition + numbers. Costs \$0.
 2. **Short-form video (TikTok + YT Shorts + Reels):** 15–30s clips — "I turned \$100k into \$480k in a 6-hour crypto contest." Reuse the existing **share/screenshot** surfaces as content fuel.
 3. **Reddit / Discord communities:** participate (not spam) in r/CryptoCurrency, r/CryptoMarkets, trading Discords; run a "community cup" using the existing 20-player 6h sprint — perfect free tournament unit.
 4. **ASO + App Store featuring:** strong screenshots + "Today" submission to Apple; contest apps with seasonal events get featured.
 5. **Crypto newsletters / micro-influencers (affiliate, not cash):** give creators a referral code; they earn Premium/XP per signup — converts paid influencer marketing into performance/free.
 6. **Push re-engagement (already built):** use `pushDeviceService` + campaign Lambda for "your sprint starts in 1h," "you dropped to #4," "league resets tonight" — keeps the retention north-star healthy with no media spend.
 7. **Launch moments:** Product Hunt + a "Season 1 Championship" launch event to concentrate signups.
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Content auto-poster bot ("HypeBot") — automate X / Instagram / TikTok

The biggest cost of the organic strategy (and *all* of Plan B) is the manual content grind. A bot removes it: on every post-worthy moment it generates the copy + hashtags + media and publishes to all three platforms. This reuses the app's **existing EventBridge + Lambda** stack (the same infra that already auto-creates a contest every 6h — `amplify/functions/create-rolling-contest/`), so it's a natural extension, not new plumbing.

Pipeline (per post)

1. **Trigger:** EventBridge fires on a real moment — contest close (`close-competition/handler.ts`), a new season champion, a big P&L (>X%), or a fixed daily slot. No fabricated hype.
2. **Fetch the moment:** query DynamoDB (`FinishedCompetition` , `CompetitionEntry` , leaderboard) for the facts — winner handle (respecting `leaderboardVisible`), P&L, contest type, # players.
3. **Generate copy + tags (Claude API):** send the moment facts to the Anthropic API (`claude-sonnet` for quality, or `claude-haiku` for cheap high-volume) with a brand-voice system prompt. Return JSON: `{ caption, platformVariants, hashtags[], altText }`. The model

writes platform-appropriate copy (punchy/emoji for TikTok & IG, tighter for X) and selects **rotating, relevant hashtags** (#crypto #cryptotrading #tradingchallenge #fantasycrypto ...) — rotation avoids spam/shadowban flags from repeated identical tag sets.

4. **Source stock media + render:** text-only won't work for IG/TikTok — they require image/video, and strong visuals lift engagement. Two layers:
 - **Fetch high-quality stock** by keyword (crypto, bitcoin, trading desk, charts, finance) — keywords chosen by the Claude step in (3) — from **licensed free stock APIs: Pexels** (images + video, free API, commercial use, no attribution), **Pixabay** (images + video, free), **Unsplash** (images, free API). Pick the best-scoring asset; cache to S3 to avoid refetching. **Do NOT scrape arbitrary web images/video** — copyright/ToS risk. Stock APIs only, and store the asset's license/source URL with each post for an audit trail.
 - **Composite the brand layer on top:** overlay the **result card** (leaderboard/P&L, the existing dark+neon theme) + logo + caption onto the stock image (Satori/ `@vercel/og`) or onto a 10–15s stock **video** background via **Remotion** for TikTok/Reels. Stock = the eye-catching backdrop; your real stats = the foreground. Store final render to S3 → public URL for the publishers.
5. **Publish** to each platform API (below).
6. **Human-in-the-loop (recommended):** post to a **review queue** (Slack/email approval) before going live, at least until voice + compliance are trusted. Avoids platform-ToS automation issues and off-brand/compliance misses.

Platform APIs & realities (important caveats)

Platform	API	Reality
X	X API v2 <code>POST /2/tweets</code> (+ media upload)	Easiest. Note write access needs a paid tier (~\$100/mo Basic) or the limited free tier — the one place a "\$0" plan may hit a wall; mitigate by posting X manually under Plan B.
Instagram	Instagram Graph API — Content Publishing (<code>/media</code> → <code>/media_publish</code>)	Free, but requires an IG Business/Creator account + linked Facebook app + review. Posts images/Reels from a hosted URL.
TikTok	TikTok Content Posting API (Direct Post)	Free, but requires an approved developer app + audited scopes; video only. Highest approval friction — expect lead time.

Implementation shape

- A standalone **Node/TypeScript** module (e.g. `scripts/hypebot/` or a new `amplify/functions/social-poster/`) with: `momentSource.ts` (DynamoDB queries), `generate.ts` (Anthropic call + JSON schema), `stock.ts` (Pexels/Pixabay/Unsplash fetch +

license capture), `render.ts` (composite card/video over stock), and one `publishers/{x,instagram,tiktok}.ts` each. Secrets (API + stock tokens) in AWS Secrets Manager / SSM, never committed.

- Config: posting cadence caps (e.g. max 3/day/platform), quiet hours, per-platform on/off, dry-run mode.
- Cost: Claude calls are pennies/post at Haiku/Sonnet; the only real cost is X write access.

Compliance & safety

- FTC: brand account is first-party, but keep claims truthful ("paper trading / no real money").
- Crypto-promo & automation policies differ per platform — the review queue + rate caps keep you inside ToS. Never auto-DM or auto-follow; this bot only **posts**.

Plan B — launch with \$0 media budget (pure organic)

A complete fallback if you want to spend **nothing** on ads. Cut Apple Search Ads and all paid creative testing; the engine becomes content + community + the referral loop. The product is already built and the costable infra (PostHog/Branch/Amplitude free tiers, Apple Sign-In) is \$0, so this is fully viable — it trades money for **founder time + consistency**.

What changes vs Plan A: remove the entire paid table. Everything in "Free / organic acquisition" becomes the whole strategy, plus:

1. **Founder-led content is the budget.** Commit to a non-negotiable cadence: 1 build-in-public post/day on X + 3 short-form videos/week (TikTok/Shorts/Reels) from real leaderboard moments. This is the single highest-leverage \$0 activity — treat it like a job, not an afterthought.
2. **Referral loop does the paid channel's job.** With no ad spend, "Recruit & Rise" (below) is your *only* scalable acquisition — prioritize building it first and make its rewards generous early.
3. **Community tournaments as distribution.** Run weekly "community cups" on the existing 20-player 6h sprint inside crypto Discords/subreddits — you bring the prize (XP/Premium codes, \$0), they bring the players. Partner with community mods, not paid influencers.
4. **Affiliate-only creators.** No cash to influencers — give them referral codes; they earn Premium/XP/"Ambassador" status per activation. Converts influencer marketing to pure performance.
5. **Earn featuring instead of buying installs.** Heavy ASO + a polished Product Hunt launch + an Apple "Today" submission tied to a "Season 1 Championship" event — concentrate organic signups into a moment.
6. **Cross-post free everywhere.** Reuse each video across TikTok/Shorts/Reels/X; repurpose top posts into a simple weekly email (Resend is already wired for transactional — extend to a digest).

Trade-offs (be honest): slower, spikier growth; depends entirely on content consistency and one or two channels catching; no fast CAC read. **Trigger to graduate to Plan A:** once any organic channel shows repeatable signups *and* D7 ≥ target, introduce the \$250/mo Apple Search Ads layer to pour fuel on what already works.

Referral program — "Recruit & Rise" (highest-leverage growth investment)

For a free, competitive, social app with weak acquisition infra, a referral loop is the best \$0-media growth lever — but it must be **built** (the untracked duel codes don't count). Reuses the existing XP/pass/league systems so it needs no real money (fits the XP-only launch).

Mechanic

- Each user gets a unique **referral code + Branch deep link** (e.g. `cryptocomp.app/r/AB12CD`).
- New user installs via link → attributed to referrer (requires Phase 0 deep-link attribution).
- All rewards gate on **activation = invitee completes their first contest**, never on raw install — protects against abuse and directly serves the retention north-star.

Two-sided reward (per successful referral)

Recipient	Reward	Source system
Invitee (new user)	+3 contest passes + 1,000 XP welcome bonus	weekly-pass grant + XP in <code>gamification.ts</code>
Referrer	+2 contest passes + 750 XP	same
Both, if invitee buys Premium within 14d	+1 free month No-Ads for referrer	RevenueCat entitlement

Passes are the natural currency (free users get 5/wk in `gamification.ts` ; 1 pass = 1 contest entry), so referrals literally buy more play — tightly coupled to engagement.

Milestone tiers (cumulative *activated* referrals → status + perks)

Tier	Activated referrals	Perks
Scout	1	Referral badge on profile
Recruiter	3	+5 passes, exclusive profile flair
Captain	10	Permanent +2 weekly passes, "Captain" league frame
Ambassador	25	Free Premium while active + early access to new contest types

Anti-abuse

- Reward only on first-contest activation + device/IP dedupe + Apple attribution; cap rewards/day.
- Self-referral and reinstall fraud blocked via Branch fingerprint + Cognito account age.

Surfaces (where the loop lives)

- **Post-win share sheet** — "You placed #2 — challenge a friend" (peak motivation moment).
- **Profile** — referral code, progress to next tier, "invite" CTA.
- **Upgrade 1v1 duel codes into tracked referral links** (small change in `CompeteScreen.tsx`) — the duel invite already implies a friend; make every duel a tracked referral.
- **Onboarding** — "have a code?" field so invitees claim their welcome bonus.

Seasonal Recruiter Cup (top-5 prize)

- **"Top Recruiters this Season"** leaderboard reusing the existing league/season UI — ranks users by **activated referrals** in the season, turning recruiting into a competition itself.
- **Top 5 recruiters each season win a prize.** Reward mode follows the same `PAYOUTS_ENABLED` / `CONTEST_CASH_PRIZES` gate as contests — **XP/perks now, cash later** — so it ships at launch with zero legal/payment surface and flips to cash with one flag, no rebuild:

Rank	XP-only mode (launch)	Cash mode (when payouts on)
1st	25,000 XP + 3 mo Premium + "Season Recruiter" frame	\$250 + 1 mo Premium
2nd	15,000 XP + 2 mo Premium	\$150
3rd	10,000 XP + 1 mo Premium	\$100
4th	6,000 XP	\$50
5th	4,000 XP	\$25

Sample cash pool **\$575/season** — modest, tunable, and well under the `$4,999` compliance cap in `amplify/functions/lib/contestCompliance.ts`.

- **Reuse, don't rebuild:** pay cash through the **same Stripe Connect payout path** as contest prizes (`close-competition` / `process-withdrawals`), and gate XP-vs-cash on the existing flag in `src/constants/featureFlags.ts`.
- **Compliance note (cash mode):** referrals stay **free actions**, so it remains a free-entry skill/sweepstakes consistent with `docs/official-contest-rules.html`; still apply the same 1099/\$600 reporting + dedupe/anti-fraud rules before enabling cash. Keep per-user seasonal cash under the 1099 threshold initially.
- Push nudge via `pushDeviceService` when a friend activates and when seasonal standings change.

Targets

- Goal: **K-factor 0.15–0.3** at launch (every 4–7 active users bring 1 more), trending up as surfaces optimize. Track invites sent → installs → activations as a funnel in analytics.

Prereq & sequencing: depends on Phase 0 deep-link attribution + analytics. Ship "Recruit & Rise" immediately after Phase 0, before scaling any paid spend — it lowers blended CAC for everything after.

Suggested timeline

- **30 days:** Phase 0 (analytics, attribution, Apple Sign-In, ASO). App approved & live. Begin organic (X build-in-public, first community cup). No paid yet.
 - **60 days:** Ship referral loop. Start Apple Search Ads (\$250/mo). Begin short-form video cadence (2–3/wk). Measure D1/D7, CAC, contest fill rate.
 - **90 days:** Double down on whichever organic channel + the one paid creative that work; kill the rest. Re-evaluate flipping select contests to cash prizes once weekly-active + fill-rate are healthy.
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Metrics & targets (north star: retained active users)

- **North star:** Weekly Active Competitors (users who join ≥ 1 contest/week).
 - **Guardrails:** D1 $\geq 35\%$, D7 $\geq 15\%$, D30 $\geq 6\%$ (free mobile-game benchmarks; iterate from real data); onboarding→first-trade completion; **6h-sprint fill rate** (are the 20 seats filling?).
 - **Acquisition:** Apple Search Ads CAC; referral K-factor (invites sent $\times$ conversion); organic vs paid mix.
 - **Don't optimize raw installs** — explicitly a guardrail, not a goal.
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Implementation workstreams (everything to build — nothing deferred)

WS1 — Analytics & attribution (Phase 0, blocking)

- **Scope:** PostHog/Amplitude funnel (install → signup → onboarding → first trade → first contest → D1/D7) + Branch deep links + URL scheme/universal links.
- **Files:** `app.json`, `App.tsx`, `OnboardingWalkthrough.tsx`, `TradeScreen.tsx`, `CompeteScreen.tsx`.
- **Dep:** none. Unblocks WS4, WS5, all paid spend.

WS2 — Auth friction reduction (Phase 0)

- **Scope:** Sign in with Apple + optional guest mode.
- **Files:** `src/screens/AuthScreen.tsx`, `amplify/auth/`. **Dep:** none.

WS3 — ASO pass (Phase 0, \$0)

- **Scope:** keywords, competition-led screenshots, US-English; Product Hunt + Apple "Today" assets.
- **Files:** App Store Connect metadata; screenshot assets. **Dep:** app approved.

WS4 — Referral "Recruit & Rise" + Seasonal Recruiter Cup

- **Scope:** codes/links, two-sided rewards, tiers, top-5 cup (XP-now/cash-later), anti-abuse, surfaces.
- **Files:** new referral service + `amplify/data/resource.ts`, `CompeteScreen.tsx`, `src/services/gamification.ts`, seasonal-close job, reuse `close-competition / process-withdrawals` + `featureFlags.ts`. **Dep:** WS1.

WS5 — HypeBot auto-poster (X / IG / TikTok, stock media)

- **Scope:** EventBridge → moment → Claude copy+hashtags → stock + render → publish (review queue).
- **Files:** new `amplify/functions/social-poster/` (`momentSource/generate/stock/render/publishers`); EventBridge rule; Secrets Manager. **Dep:** pairs with WS1. Note: X write tier (~\$100/mo).

WS6 — Real-money prize enablement (trigger-gated)

- **Scope:** flip `PAYOUTS_ENABLED / CONTEST_CASH_PRIZES` for select contests + Recruiter Cup once retention/fill-rate healthy. Sweepstakes registration, 1099 (\$600+), Stripe QA, prize-cap enforcement.
- **Files:** `featureFlags.ts`, `WithdrawScreen.tsx`, `contestCompliance.ts`, `stripe-webhook / process-withdrawals`, `docs/official-contest-rules.html`.
- **Dep:** WS1 + legal review. **Trigger:** DAU/fill-rate threshold, not a date.

WS7 — International / non-US expansion

- **Scope:** localize ASO, vet per-market crypto/prize ad rules, geo-gate cash contests.
- **Files:** store metadata; `contestCompliance.ts`. **Dep:** healthy US retention + legal review.

Suggested build order

WS1 + WS2 + WS3 (Phase 0) → WS4 + WS5 (growth engine) → WS6 (monetize once retained) → WS7 (scale geo).

Verification (end-to-end)

- **WS1:** fire a test event end-to-end; confirm the funnel appears in analytics and a Branch link opens the app to the right screen with attribution recorded.
- **WS2:** complete Apple Sign-In + guest flow on a TestFlight build; a guest can reach a contest and later bind an account.
- **WS4:** install via a referral link on a clean device → complete first contest → confirm both users' rewards and that Recruiter Cup standings update at season close (XP mode).
- **WS5:** run the poster in **dry-run**; confirm copy/hashtags + composited stock render look right and land in the review queue before any live post.
- **WS6:** in staging, flip the flag and run a contest + cup settlement through Stripe test mode; verify payout records, the \$4,999 cap, and 1099 thresholds.
- Gestures/visuals and on-device flows can't be verified headlessly — build to TestFlight and test on a real device.